6. Statistical Inference: Significance Tests

Goal: Use statistical methods to test hypotheses such as

"Mental health tends to be better at higher levels of socioeconomic status (SES)"

"For treating anorexia, cognitive behavioral and family therapies have same effect."

Hypotheses: Predictions about a population expressed in terms of parameters

A **significance test** uses data to summarize evidence about a hypothesis by comparing sample estimates of parameters to values predicted by the hypothesis.

We answer a question such as, “If the hypothesis were true, would it be unlikely to get estimates such as we obtained?”

Five Parts of a Significance Test

- **Assumptions** about type of data (quantitative, categorical), sampling method (random), population distribution (binary, normal), sample size (large?)

- **Hypotheses**:

Null hypothesis (H_0): A statement that parameter(s) take specific value(s) (Often: “no effect”)

Alternative hypothesis (H_a): states that parameter value(s) in some alternative range of values

p. 1 examples?

- **Test Statistic**: Compares data to what null hypo. H_0 predicts, often by finding the number of standard errors between sample estimate and H_0 value of parameter
- ***P*-value (*P*):** A probability measure of evidence about H_0, giving the probability (under presumption that H_0 true) that the test statistic equals observed value or value even more extreme in direction predicted by H_a.
  - The smaller the *P*-value, the stronger the evidence against H_0.
- **Conclusion**:
  - If no decision needed, report and interpret P-value

- If decision needed, select a cutoff point (such as 0.05 or 0.01) and reject H_0 if P-value ≤ that value

- The most widely accepted minimum level is 0.05, and the test is said to be *significant at the .05 level* if the P-value ≤ 0.05.

- If the *P*-value is not sufficiently small, we fail to reject H_0 (then, H_0 not necessarily true, but it is plausible)

- Process is analogous to American judicial system
  - H_0: Defendant is *innocent*
  - H_a: Defendant is *guilty*

Significance Test for Mean

- *Assumptions*: Randomization, quantitative variable, normal population distribution
- *Null Hypothesis*: H_0: $\mu = \mu_0$ where μ_0 is particular value for population mean

  (typically no effect or change from a standard)
- *Alternative Hypothesis*: H_a: $\mu \neq \mu_0$

  (*2-sided* alternative includes both > and <)
- *Test Statistic*: The number of standard errors the sample mean falls from the H_0 value

$$t = \frac{\bar{y} - \mu_0}{se} \quad \text{where} \quad se = s / \sqrt{n}$$

When H_0 is true, the sampling dist of the t test statistic is the t distribution with $df = n - 1$.

- *P-value*: Under presumption that H_0 true, probability the test statistic equals observed value or even more extreme (i.e., larger in absolute value), providing stronger evidence against H_0
  - This is a *two-tail* probability, for the two-sided H_a

- *Conclusion*: Report and interpret *P*-value. If needed, make decision about H_0

Example: Anorexia study (revisited)

- Weight measured before and after period of treatment

- y = weight at end – weight at beginning

- In previous chapter, we found CI for population mean of y based on n=17 girls receiving "family therapy," with data

y = 11.4, 11.0, 5.5, 9.4, 13.6, -2.9, -0.1, 7.4, 21.5, -5.3, -3.8, 13.4, 13.1, 9.0, 3.9, 5.7, 10.7

Is there evidence that family therapy has an effect?

- Let μ = population mean weight change
- Test H_0: μ = 0 (no effect) against : μ ≠ 0.
- Data have

Variable	N	Mean	Std.Dev.	Std. Error Mean
weight_change	17	7.265	7.157	

Recall that *se* obtained as

$$se = s / \sqrt{n} =$$

- Test Statistic *(df = ??):*

$$t = \frac{\bar{y} - \mu_0}{se} =$$

- *P*-Value:

Note *t* table (Table B, p. 593) tells us $P(t > 3.686) = 0.001$, so test statistic $t = 3.686$ would have P-value = 0.002

If H_0 were true, prob would = of getting sample mean at least 4.2 standard errors from null value of 0.

•Conclusion: Very strong evidence that the population mean differs from 0. (Specifically, it seems that $\mu > 0$, as also suggested by 95% CI of (3.6, 10.9) found in Chap. 5 notes)

SPSS output

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
weight_change	17	7.265	7.1574	1.7359

One-Sample Test

	Test Value = 0					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
weight_change	4.185	16	.001	7.2647	3.58	10.945

Equivalence between result of significance test and result of CI

- When *P*-value ≤ 0.05 in two-sided test, 95% CI for μ does not contain H_0 value of μ (such as 0)
- When *P*-value > 0.05 in two-sided test, 95% CI necessarily contains H_0 value of μ

(This is true for “two-sided” tests)

- CI has more information about actual value of μ

Example: Suppose sample mean = 7.26, s = 7.16, based on n = 4

Then,

$$se = s / \sqrt{n} =$$

$$\text{and } t =$$

With $df = 3$, this has two-sided P-value = .

Not very strong evidence against null hypo.
It is plausible that μ = 0.

95% CI is (-4.2, 18.6), which contains 0
(in agreement with test result)

One-sided test about mean

Example: If study predicts family therapy has *positive* effect, could use $H_{a:}$ $\mu > 0$

Data support this hypothesis if t far out in right tail, so P-value = right-tail prob.

- P-Value: $P = P(t > 2.0) = 0.07$ (for $n = 4$ case)

For $H_{a:}$ $\mu < 0$, P-value = left-tail prob.

- P-value: $P = P(t < 2.0) = 0.93$

In practice, two-sided tests are more common

Making a decision:

The α-level is a fixed number, also called the *significance level,* such that

if *P*-value $\leq \alpha$, we "reject H_0"
If *P*-value $> \alpha$, we "do not reject H_0"
Note: We say "Do not reject H_0" rather than "Accept H_0" because H_0 value is only one of many plausible values.

Example (*n* = 4, two-sided)*:* Suppose $\alpha = 0.05$. Since *P*-value = 0.14, we do not reject H_0. But 0 is only one of a range of plausible values exhibited in 95% CI of (-4.2, 18.6).

Effect of sample size on tests

- With large *n* (say, *n* > 30), assumption of normal population distribution not important because of Central Limit Theorem.
- For small *n,* the *two-sided t* test is robust against violations of that assumption. One-sided test is *not* robust.
- For a given observed sample mean and standard deviation, the larger the sample size *n*, the larger the test statistic (because *se* in denominator is smaller) and the smaller the *P*-value. (i.e., we have more evidence with more data)
- We're more likely to reject a false H_0 when we have a larger sample size (the test then has more "power")
- With large *n*, "statistical significance" not the same as "practical significance."

Example: Suppose anorexia study for weight change had

$$\bar{y} = 1.0, s = 2.0, \text{ for } n = 400$$

$$\text{Then } se = 2.0 / \sqrt{400} = 0.1,$$

Test stat

$$t = (1.0 - 0) / 0.1 = 10.0,$$

$$P - \text{value} = 0.000000.....$$

95% CI is 1.0 ±

This shows there is a positive effect, but it is very small in practical terms.

Significance Test for a Proportion π

- Assumptions:
  - Categorical variable
  - Randomization
  - Large sample (but two-sided ok for nearly all *n*)
- Hypotheses:
  - Null hypothesis: H_0: $\pi = \pi_0$
  - Alternative hypothesis: H_a: $\pi \neq \pi_0$ (2-sided)
  - H_a: $\pi > \pi_0$ H_a: $\pi < \pi_0$ (1-sided)
  - (choose before getting the data)

- Test statistic:

$$z = \frac{\hat{\pi} - \pi_0}{\sigma_{\hat{\pi}}} = \frac{\hat{\pi} - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}$$

As in test for mean, test statistic has form

(estimate of parameter – null hypo. value)/(standard error)

= no. of standard errors the estimate falls from null hypo. value

- *P*-value:

  H_a: $\pi \neq \pi_0$ P = 2-tail prob. from standard normal dist.
  H_a: $\pi > \pi_0$ P = right-tail prob. from standard normal dist.
  H_a: $\pi < \pi_0$ P = left-tail prob. from standard normal dist.

- Conclusion: As in test for mean (e.g., reject H_0 if *P*-value ≤ α)

Example: Can dogs smell cancer? (*British Medical Journal*, Sept. 25, 2004)

- Each trial, one bladder cancer urine sample placed among six control urine samples
- Do dogs make the correct selection better than with random guessing?
- In 54 trials, dogs made correct selection 22 times.

Let π = probability of correct guess, for particular trial

H_0: $\pi = 1/7$ (= 0.143, no effect), H_a: $\pi > 1/7$

Sample proportion = 22/54 = 0.407

Standard error

$$se_0 =$$

Test statistic

$$z =$$

P-value =

There is extremely strong evidence that dogs' selections are better than random guessing

For standard α cut-off such as 0.05, we reject H_0 and conclude that $\pi > 1/7$.

Caveat: As in most medical studies, subjects were a *convenience sample.* We can not realistically randomly sample bladder cancer patients or dogs for the experiment.

Even though samples not random, important to employ randomization in experiment, in placement of bladder cancer patient's urine specimen among the 6 control specimens.

Decisions in Tests

- α-level (significance level): Pre-specified "hurdle" for which one rejects H_0 if the P-value falls **below** it. (Typically 0.05 or 0.01)

P-Value	H_0 Conclusion	H_a Conclusion
$\leq .05$	Reject	Accept
$> .05$	Do not Reject	Do not Accept

- Rejection Region: Values of the test statistic for which we reject the null hypothesis
  - For 2-sided tests with $\alpha = .05$, we reject H_0 if $|z| \geq 1.96$

Error Types

- Type I Error: Reject H_0 when it is true
- Type II Error: Do not reject H_0 when it is false

Test Result –	Reject H_0	Don't Reject H_0
True State		
H_0 True	Type I Error	Correct
H_0 False	Correct	Type II Error

P(Type I error)

- Suppose α-level = 0.05. Then, P(Type I error) = P(reject null when true) = $P(|z| > 1.96) = 0.05$
- i.e., the α-level is the P(Type I error).
- Since we "give benefit of doubt to null," traditional to take α to be small, usually 0.05 but 0.01 to be very cautious not to reject null when it may be true.
- As in CIs, don't make α too small since as α goes down, β = P(Type II error) goes up

  (Think of analogy with courtroom trial)
- Better to report *P*-value than merely whether reject H_0

  (make sure you do homework 6.24)

P(Type II error)

- P(Type II error) = β depends on the true value of the parameter (from the range of values in H_a).
- The farther the true parameter value from the null value, the easier it is to reject null, and P(Type II error) goes down.
- Power of test = 1 - β = P(reject null when false)
- In practice, you want a large enough *n* for your study so that *P*(Type II error) is small for the size of effect you expect.

Example: Testing new treatment for anorexia

For a new treatment, we expect a mean weight change of about 10 pounds, with standard deviation about 10. If our study will have 20 subjects, approximately what is P(Type II error) if we test H_0: μ = 0 against H_a: μ > 0, using α = 0.05?

- We fail to reject H_0: μ = 0 if we get *P*-value > 0.05
- We get *P*-value = 0.05 if test statistic *t* = (i.e., with *df* = 19, 0.05 is right-tail prob. above , so "rejection region" is values of *t* >)
- With *n* = 20, the standard error should be about

$$se =$$

- We get t = 1.729 if the sample mean is about

- So, we'll get t < 1.729 and P-value > 0.05 (and make Type II error) if we get sample mean < .

- But, if actually μ = 10, a sample mean of 3.87 is about
  standard errors from μ
  (i.e., std. errors below μ = 10)

- When df = 19, the probability falling at least 2.74 standard errors below the mean is about 0.007. So, there's little chance of making a Type II error.

- But what if μ actually is only 5? (exercise)

Limitations of significance tests

- *Statistical significance* does not mean *practical significance*
- Significance tests don’t tell us about the *size* of the effect (like a CI does)
- Some tests may be “statistically significant” just by chance (and some journals only report “significant” results)

Example: Are many medical "discoveries" actually Type I errors?

Reality: Most medical studies are "non-significant," not finding an effect.

In medical research, when effects exist but are not strong, they may not be detected with the sample size practical for a study.

(A *British Medical Journal* article in 2001 estimated that when an effect truly exists, P(Type II error) = 0.50!)

In medical studies, suppose an effect actually exists 8% of the time. Could a substantial percentage of medical "discoveries" (i.e., significant results) actually be Type I errors?

Simple-minded solution: Draw a ***tree diagram*** to show what we'd expect to happen with many studies (say, 1000)

```
                 True                    Decision
                 effect                  Reject null?

                                            Yes (  )
                 Yes (  ) ---------------|
                                            No (  )
1000 studies---|
                                            Yes (   )
                 No (   ) ----------------|
                                            No (   )
```

Of rejected nulls, Type I error rate = 46/(46+40) = 0.53

Moral of the story: Be skeptical when you hear reports of new medical advances.

There may be *no* actual effect (i.e. the entire study may merely be a Type I error!)

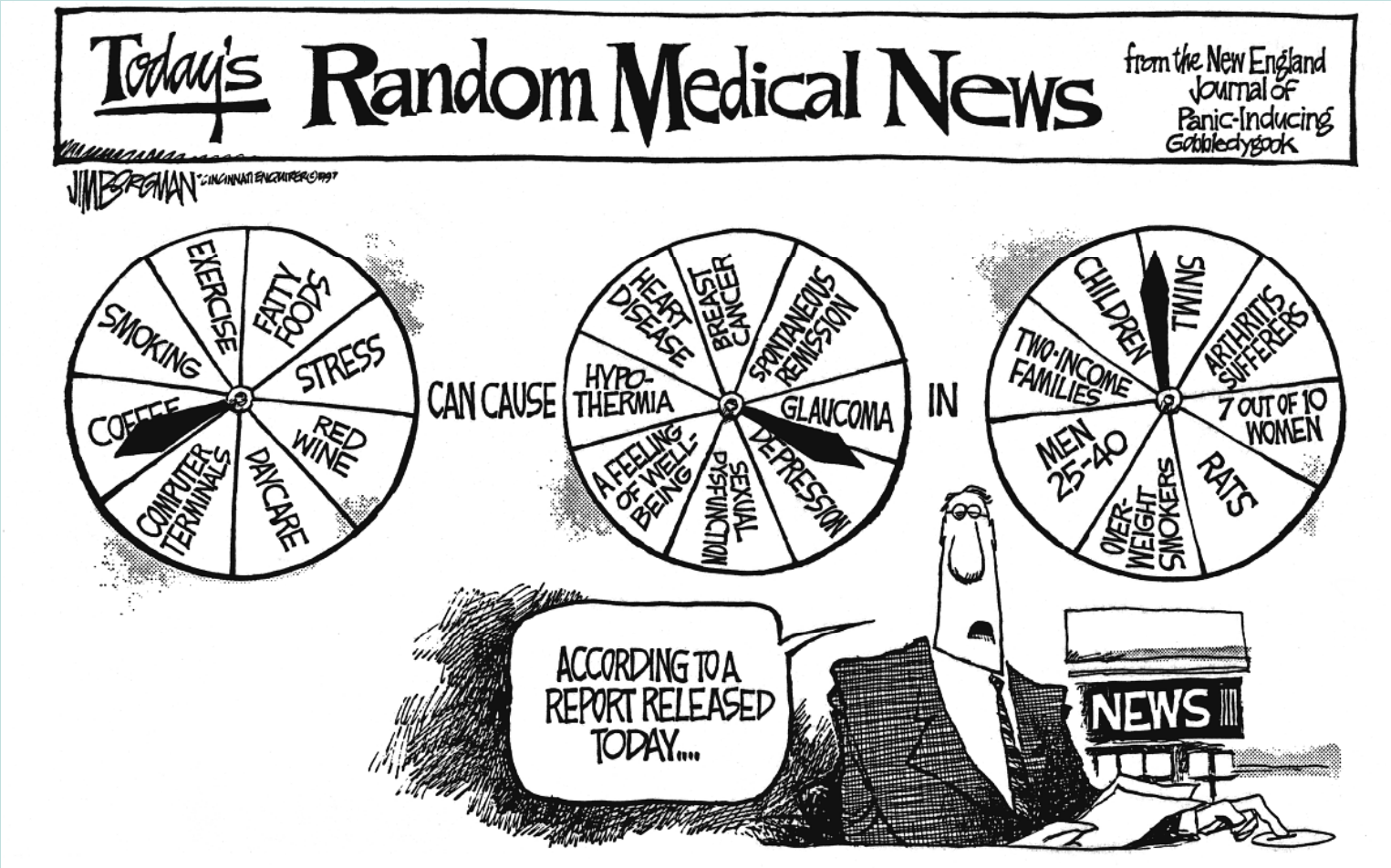
If actual effect exists, we may be seeing a sample outcome in right-hand tail of sampling distribution of possible sample effects, and the actual effect may be much weaker than reported.

(picture of what I mean by this)

Actual case: A 1993 study (with a small sample) estimated that injections of magnesium could double the chance of surviving a major heart attack.

A much larger later study of 58,000 heart attack patients found no effect at all.

Figure from Agresti and Franklin, *Statistics: The Art and Science of Learning from Data* (p. 468)



The binomial distribution

If

- Each *observation* is binary (one of two categories)
- Probabilities π for category 1, 1-π for category 2 for each observation
- Observations are independent

then for *n* observations, the number *x* in category 1 has

$$P(x) = \frac{n!}{x!(n-x)!}\pi^x(1-\pi)^{n-x}, \quad x = 0,1,\ldots,n$$

This can be used to conduct tests about π when *n* is too small to rely on large-sample methods (e.g., when expected no. observations in either category < about 10)

Example: Exercise 6.33 (ESP)

- Person claims to be able to guess outcome of coin flip in another room correctly more often than not

- π = probability of correct guess (for any one flip)

- H_0: $\pi = 0.50$ (random guessing)
- H_a: $\pi > 0.50$ (better than random guessing)

- Experiment: $n = 5$ trials, $x = 4$ correct

- Under H_0, prob. of 4 correct out of 5 is

$$P(4) = \frac{n!}{x!(n-x)!}\pi^x(1-\pi)^{n-x} =$$

- Probability of the more extreme result of 5 correct is

$$P(5) =$$

- For H_a : $\pi > 0.50$, *P*-value is probability of observed result or result even more extreme = $P(4) + P(5) =$. There is not much evidence to support the claim.

Notes about binomial distribution

- Binomial is the most important probability distribution for categorical data
- Binomial dist for *x* = number in category 1 has

$$\mu = E(x) = n\pi, \quad \sigma = \sqrt{n\pi(1-\pi)}$$

whereas sample proportion $\hat{\pi}$ = *x/n* has

$$E(\hat{\pi}) = \pi, \quad \sigma_{\hat{\pi}} = \sqrt{\pi(1-\pi)/n}$$

Example: Poll results for proportion with n = 1000, π = 0.50

- x = *number* in category of interest has

$$\mu = E(x) = n\pi = 1000(0.50) = 500, \quad \sigma = \sqrt{n\pi(1-\pi)} = \sqrt{1000(0.50)(0.50)} = 15.8$$

- *proportion* $\hat{\pi}$ in category of interest has

$$E(\hat{\pi}) = \pi = 0.50, \quad \sigma_{\hat{\pi}} = \sqrt{\pi(1-\pi)/n} = \sqrt{(0.50)(0.50)/1000} = 0.0158$$

(effect of n? Compare to n = 10)